Askar Shamshidinov

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AI engineer with two years of web development before AI. I build **computer vision and LLM features that work in real services**: YOLO11 detection and DeepSORT tracking on live CCTV (best mAP50 0.81), a production RAG assistant on a Telegram bot handling 50–300 queries a day, a tool-calling agent measured at 8–13% → 96–100% accuracy, and embedding fine-tuning for Kyrgyz retrieval (Accuracy@1 0.56 → 0.74). Python, PyTorch, FastAPI, React / Next.js / NestJS, PostgreSQL. English (advanced), Russian and Kyrgyz (native), Korean (conversational).

PROJECTS

Korea Visa Guide — tool-calling LLM agent (mobile app) 2026

TypeScript, Expo / React Native, function calling, Gemini / DeepSeek / Claude APIs, Tavily web search

- Rebuilt a search-only chatbot as an agent with four tools, so dates and fees are computed by code instead of guessed by the model.
- Measured both versions on 24 checkable questions over three runs: accuracy 8–13% → 96–100%.

Smart Traffic Monitoring — detection and counting on live CCTV Mar. – May 2025

Python, YOLO11, DeepSORT, PyTorch, OpenCV, FastAPI · team project with PiaSpace

- Owned the AI part: dataset, 4 YOLO11 training experiments on a Linux GPU server (best mAP50 0.81), DeepSORT tracking and line-crossing counts; integrated the model into the team's FastAPI backend.

Document AI & OCR — receipts and passports

Python, LayoutLMv3, OCR, OpenCV

- Compared a text-only model with LayoutLMv3 (text + layout) for receipt field extraction and tested when results can be accepted automatically.
- Built OCR post-processing for passports that decides which results are safe to accept by cross-checking MRZ check digits and line geometry.

Kyrgyz Semantic Retrieval Research Sept. 2026 – Present

Python, PyTorch, sentence-transformers, Hugging Face

- Error analysis found an evaluation flaw (34% of gold passages duplicated as distractors) that understated all models by up to 26 points and reversed the BM25-vs-neural conclusion.
- Fine-tuned multilingual-e5-base with mined hard negatives: Accuracy@1 on human-written questions 0.56 → 0.74; compared 9 systems on 3 test sets with bootstrap CIs and Kendall's τ.

EXPERIENCE

AI Developer Intern — Motion Group Apr. – Aug. 2026

Kyrgyz Republic

- Built a RAG assistant over 105 internal documents (~600 pages): document parsing, chunking, embeddings and vector retrieval, answers grounded in the retrieved sources.
- Launched it as a Telegram bot used by employees every day (50–300 queries a day) and kept it running in production.

Frontend Developer — MotionWeb Sept. 2021 – Oct. 2023

Russia (client projects under NDA)

- Built responsive frontends in React, TypeScript, HTML and CSS that adapt to different screen sizes; integrated REST APIs.
- Built the frontend of a Web3 token launchpad: MetaMask wallet connection and smart-contract calls via Ethers.js.

Full-Stack Developer Intern — NEVIS TECH ITDB Jan. – Apr. 2021

Dubai, UAE

- Built CRM systems and admin panels: React / Next.js frontends and NestJS backends with authentication and REST endpoints.

TECHNICAL SKILLS

LLM & agents: function calling / tool use, RAG, hybrid retrieval (vector + BM25), reranking, embeddings, evaluation sets, Gemini / Claude / DeepSeek APIs, Ollama
Languages & web: Python, TypeScript, JavaScript, React, Next.js, React Native, NestJS, FastAPI, REST APIs, PostgreSQL
ML & vision: PyTorch, YOLO11, DeepSORT, OpenCV, SAHI, LayoutLMv3, OCR, sentence-transformers, Hugging Face
Tools: Git, Linux, Docker, pytest

EDUCATION

Woosong University, Endicott College of International Studies — B.S. in Artificial Intelligence & Big Data 2022 – Aug. 2027 (expected)

Daejeon, South Korea. Available full-time from December 2026 (university co-op internship or student work permit; no visa sponsorship needed).

LANGUAGES

English (advanced) · Russian (native) · Kyrgyz (native) · Korean (conversational, studying; working language English)